

Week 2: Nodal and Mesh Analysis

Node voltage method, supernodes, mesh current method, supermeshes, dependent sources, choosing a method

In-depth notes covering all core material of this chapter: theory, engineering shortcuts, exam traps, 9 worked examples, and 8 practice problems with full solutions.

Based on: Rizzoni & Kearns, *Principles and Applications of Electrical Engineering*, 7th ed., §1.7–1.9

Inspired by: ELEC 275, Principles of Electrical Engineering, Concordia University

By **Vladislav Shkuratov**, made with Claude (AI). Independent study notes, not affiliated with Concordia University or any instructor.

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How to use this deck

- **“Try it” slides:** the problem alone, with empty space. Do it first.
- **Next slide:** the clean solution, step by step.
- **Purple boxes:** engineering shortcuts, the fast way to see or solve things.
- **Orange boxes:** traps. This is where marks get lost on exams.
- Formulas are tagged **[on formula sheet]** or **[NOT on formula sheet]**, based on the sheet given in ELEC 275 exams.
- **End of deck:** final-check problems, then a full answer sheet.

The right side of each slide is left empty for your notes.

Why we need systematic methods

The problem

Week 1 tricks (series, parallel, dividers) fail as soon as a circuit has several sources or a resistor that's neither series nor parallel with anything.

The fix: two recipes that always work

Node voltage method (nodal): unknowns are node voltages, equations are KCL.

Mesh current method (mesh): unknowns are mesh currents, equations are KVL.

Both turn any circuit into N equations in N unknowns. You solve the system, then get every current, voltage and power from the result.

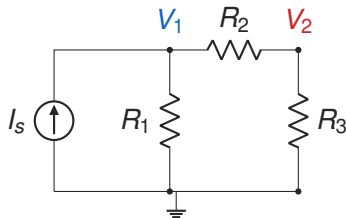
Symbols for this week

Symbol	Meaning
$V_1, V_2, \dots$	node voltage: voltage of a node measured relative to the reference node
reference node	the node chosen as 0 V (ground symbol), usually the one with most connections
$I_1, I_2, \dots$	mesh current: a current circulating around one mesh (window), clockwise
$G = 1/R$	conductance in siemens (S), makes nodal equations cleaner
n	number of nodes, including the reference
supernode	a floating voltage source plus its two nodes, treated as one big node
supermesh	two meshes merged around a shared current source

Node voltage method: the recipe

[NOT on formula sheet: memorize]

- 1 Pick the reference node (0 V), usually the bottom wire.
- 2 Label every other node: $V_1, V_2, \dots$
- 3 Write KCL at every non-reference node, with currents written using node voltages.
- 4 Solve the $n - 1$ equations.
- 5 Get anything else (branch currents, power) from the node voltages.



Why nodal analysis always works

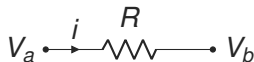
- 1. Node voltages are enough.** Once you know every node voltage, every element's voltage is a difference $V_a - V_b$, and every resistor current follows from Ohm's law. So the node voltages are the only real unknowns.
- 2. One equation per unknown.** KCL gives one equation per node. With n nodes, one is the reference (known, 0 V), so there are $n - 1$ unknowns and $n - 1$ KCL equations.
- 3. The reference's KCL adds nothing.** Its KCL is the sum of all the others, so writing it only repeats information.

Mesh is the mirror image

Mesh currents are enough to find every branch current (a branch carries one mesh current, or the difference of two). KVL gives one equation per mesh, and meshes are automatically independent.

Writing currents with node voltages

[NOT on formula sheet: memorize]



$$i = \frac{V_a - V_b}{R}$$

Current flowing from a to b is “this node minus that node, over R ”.

Engineering shortcut: “everything leaving”

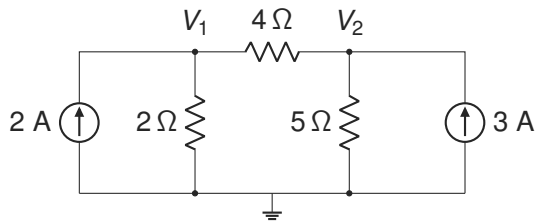
At each node, write every resistor current as **leaving**: $(V_{\text{this}} - V_{\text{other}})/R$.

Then set

$$\sum (\text{resistor currents leaving}) = \sum (\text{source currents entering}).$$

No guessing directions, and the signs come out right every time.

Try it: Example 1, nodal with two nodes



Use nodal analysis to find V_1 and V_2 .

Write the equations with variables first.

your work here ↓

Example 1: solution (1/2)

Plan two unknown nodes, only current sources: KCL at each, with every resistor current written as *leaving* the node

KCL at V_1 leaving through the $2\ \Omega$ ($V_1/2$) and the $4\ \Omega$ ($(V_1 - V_2)/4$)
= the 2 A entering: $\frac{V_1}{2} + \frac{V_1 - V_2}{4} = 2$

Clear $\times 4$: $2V_1 + V_1 - V_2 = 8 \Rightarrow 3V_1 - V_2 = 8$

KCL at V_2 leaving through the $4\ \Omega$ and the $5\ \Omega$ = the 3 A entering:
 $\frac{V_2 - V_1}{4} + \frac{V_2}{5} = 3 \xrightarrow{\times 20} -5V_1 + 9V_2 = 60$

Example 1: solution (2/2)

Substitute first equation: $V_2 = 3V_1 - 8$. Then
 $-5V_1 + 27V_1 - 72 = 60 \Rightarrow 22V_1 = 132$

Result

$$V_1 = 6 \text{ V}$$

$$V_2 = 3(6) - 8 = 10 \text{ V}$$

Check

at V_1 : $3 + (-1) = 2 \text{ A}$ ✓ at V_2 : $1 + 2 = 3 \text{ A}$ ✓

Shortcut: nodal matrix by inspection

Engineering shortcut: conductance matrix

(Circuits with only resistors and current sources.)

Diagonal G_{kk} = sum of all conductances touching node k .

Off-diagonal G_{kj} = **minus** the conductance directly between nodes k and j .

Right side = total current-source current **entering** node k .

Example 1 straight from the drawing:
$$\begin{bmatrix} \frac{1}{2} + \frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{4} + \frac{1}{5} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Always symmetric. If yours isn't, there's a sign mistake (unless there's a dependent source).

Why the by-inspection matrix works

Write KCL at node k with every resistor current leaving: each resistor from k to node j gives $G(V_k - V_j)$.

Collect the V_k terms: every resistor touching node k contributes $+G$ to V_k 's coefficient, which is the **diagonal**.

Collect the V_j terms: the resistor between k and j contributes $-G$ to V_j 's coefficient, which is the **off-diagonal**.

Sources entering node k go on the right side.

The same resistor appears as $-G$ in row k (column j) and in row j (column k): that's why the matrix is **symmetric**. A dependent source adds a term in some other column only, which breaks the symmetry.

Solving 2×2 systems fast

Cramer's rule

For $aX + bY = e$, $cX + dY = f$:

$$X = \frac{ed - bf}{ad - bc} \quad Y = \frac{af - ec}{ad - bc}$$

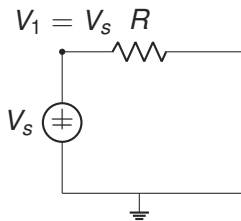
Example 1 ($3V_1 - V_2 = 8$, $-5V_1 + 9V_2 = 60$):

$$\det = (3)(9) - (-1)(-5) = 22, \quad V_1 = \frac{(8)(9) - (-1)(60)}{22} = \frac{132}{22} = 6 \checkmark$$

Calculator

For 3×3 and bigger, use your calculator's equation solver or matrix mode. Practice it before the exam. It saves 5–10 minutes per question.

Voltage source to the reference: a free node



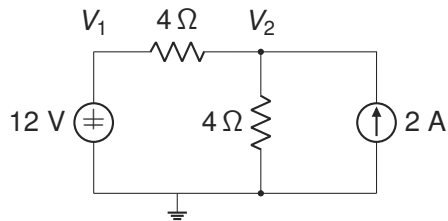
If a voltage source connects a node straight to the reference, that node's voltage is **known**: $V_1 = V_s$.

One fewer unknown, and you do **not** write KCL there. (You couldn't anyway: the source's current is unknown.)

Trap

Writing KCL at a node with a voltage source adds a new unknown (the source's current). Use $V = V_s$ instead.

Try it: Example 2, voltage source to ground



Find V_2 and the current delivered by the 12 V source.

your work here ↓

Example 2: solution

Known node $V_1 = 12 \text{ V}$ (source to ground)

KCL at V_2 $\frac{V_2 - 12}{4} + \frac{V_2}{4} = 2 \xrightarrow{\times 4} 2V_2 - 12 = 8$

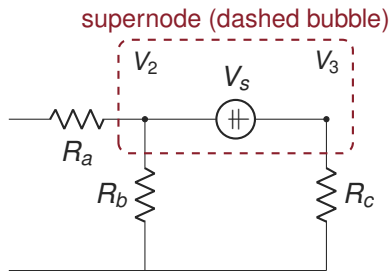
Result $V_2 = 10 \text{ V}$

Source current through the 4Ω : $\frac{12 - 10}{4} = 0.5 \text{ A out of the } +$

A voltage source's current comes **after** solving: use KCL at its node with the answers you already have.

Floating voltage source: the supernode

[NOT on formula sheet: memorize]



A voltage source between two non-reference nodes is **floating**.

1. Draw a bubble around the source and both its nodes.
2. Write **one** KCL for the whole bubble (all currents leaving it).
3. Add the source equation:
 $V_3 - V_2 = V_s$.

Trap

Don't include the voltage source's own current in the supernode KCL. It's inside the bubble, so it cancels out.

Why the supernode works

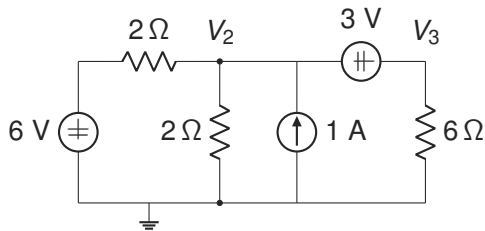
The problem: a floating source's current is unknown, and you can't write it with node voltages ($i \neq \Delta V/R$ for a source). So KCL at V_2 alone, or at V_3 alone, would add a new unknown.

The fix: draw the bubble around both nodes. That current flows *inside* the bubble (out of one node, into the other), so it cancels in the bubble's KCL. You get one clean equation.

The count: you lost one equation (two nodes, one KCL), so the source gives it back: $V_3 - V_2 = V_s$. Two unknowns, two equations.

Supermesh is the same idea for mesh: a shared current source's voltage is unknown, so walk *around* it (outer loop), and add the source equation $I_2 - I_1 = I_s$.

Try it: Example 3, supernode



The 3 V source has + on the right (V_3 side).

Find V_2 and V_3 .

your work here ↓

Example 3: solution

Known node $V_1 = 6 \text{ V}$

Source eq. $V_3 = V_2 + 3$

Supernode KCL leaving: $\frac{V_2 - 6}{2} + \frac{V_2}{2} - 1 + \frac{V_3}{6} = 0$

Substitute $\times 6$: $3V_2 - 18 + 3V_2 - 6 + V_2 + 3 = 0 \Rightarrow 7V_2 = 21$

Result

$$V_2 = 3 \text{ V},$$

$$V_3 = 6 \text{ V}$$

The 1 A source pushes current **into** the bubble, so it's -1 in a “sum of currents leaving” equation.

Dependent sources in nodal analysis

The two-step rule

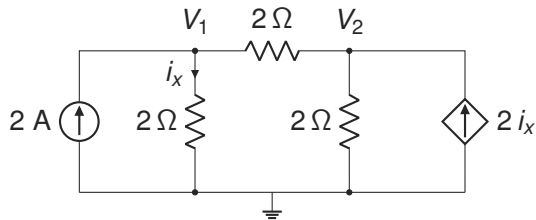
1. Write KCL as usual, treating the dependent source like a normal source.
2. Write the control variable (i_x or v_x) **in terms of node voltages**, and substitute.

Example: if i_x is the current down through a resistor R from node V_1 , then $i_x = V_1/R$.

Trap

With a dependent source, the matrix is **no longer symmetric**. The by-inspection shortcut doesn't apply. Write each KCL by hand.

Try it: Example 4, nodal with a CCCS



The dependent source pushes $2i_x$ **up** into V_2 .

Find V_1 , V_2 and i_x .

your work here ↓

Example 4: solution (1/2)

Plan KCL at both nodes, treating $2i_x$ like a known current source, then write i_x in node voltages

KCL at V_1 leaving through both 2Ω 's = 2 A entering:
$$\frac{V_1}{2} + \frac{V_1 - V_2}{2} = 2 \Rightarrow 2V_1 - V_2 = 4$$

Control i_x flows down the left 2Ω from V_1 to ground: $i_x = V_1/2$

KCL at V_2 leaving through its two resistors = the $2i_x$ entering:
$$\frac{V_2 - V_1}{2} + \frac{V_2}{2} = 2i_x = V_1$$

Example 4: solution (2/2)

Simplify $\times 2: 2V_2 - V_1 = 2V_1 \Rightarrow V_2 = 1.5 V_1$

Substitute $2V_1 - 1.5V_1 = 4 \Rightarrow V_1 = 8 \text{ V}, V_2 = 12 \text{ V},$

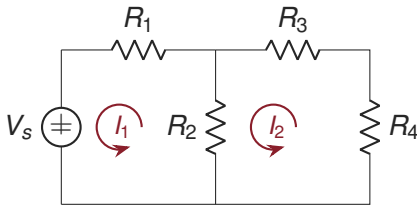
$i_x = 4 \text{ A}$

Check at $V_2: \frac{12-8}{2} + \frac{12}{2} = 8 = 2i_x \checkmark$

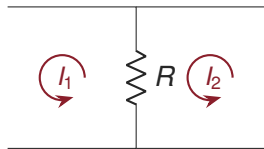
Mesh current method: the recipe

[NOT on formula sheet: memorize]

- 1 Check the circuit is flat (no crossing wires).
- 2 Draw a clockwise current in every mesh (window): $i_1, i_2, \dots$
- 3 Write KVL around every mesh.
- 4 Solve.
- 5 Branch currents = mesh current, or the **difference** of two on a shared branch.



The shared resistor



In **mesh 1's** KVL, the shared R drops $R(I_1 - I_2)$.

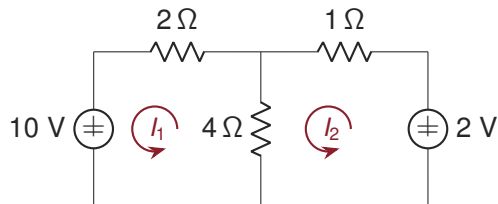
In **mesh 2's** KVL, it drops $R(I_2 - I_1)$.

Rule: **your own mesh current minus the neighbour's.**

Engineering shortcut

Walking around mesh k in its own direction, every resistor gives $-R(I_k - I_{\text{neighbour}})$. With all meshes clockwise, this never fails.

Try it: Example 5, two meshes



Both sources have + on top.

Find I_1 , I_2 and the current in the $4\ \Omega$.

your work here ↓

Example 5: solution (1/2)

Plan two meshes, only voltage sources: KVL around each, walking clockwise

Mesh 1 up the 10 V (+10), across the $2\ \Omega$ ($-2I_1$), down the shared $4\ \Omega$ where the net current is $I_1 - I_2$ ($-4(I_1 - I_2)$):
 $6I_1 - 4I_2 = 10$

Mesh 2 across the $1\ \Omega$ ($-I_2$), down the 2 V entering its + (-2), up the shared $4\ \Omega$ ($-4(I_2 - I_1)$): $-4I_1 + 5I_2 = -2$

Example 5: solution (2/2)

Cramer $\det = (6)(5) - (-4)(-4) = 14,$
$$I_1 = \frac{(10)(5) - (-4)(-2)}{14} = 3,$$
$$I_2 = \frac{(6)(-2) - (10)(-4)}{14} = 2$$

Result

$$I_1 = 3 \text{ A}$$

$$I_2 = 2 \text{ A}$$

4Ω shared branch: $I_1 - I_2 = 1 \text{ A downward}$ (positive, so in I_1 's direction)

Shortcut: mesh matrix by inspection

Engineering shortcut: resistance matrix

(Only resistors and voltage sources, all meshes clockwise.)

Diagonal R_{kk} = sum of all resistances around mesh k .

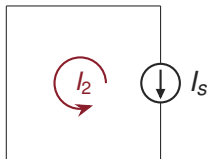
Off-diagonal R_{kj} = **minus** the resistance shared by meshes k and j .

Right side = sum of source voltage **rises** going clockwise around mesh k .

Example 5 straight from the drawing:

$$\begin{bmatrix} 2 + 4 & -4 \\ -4 & 4 + 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 10 \\ -2 \end{bmatrix}$$

Current source on the edge: a known mesh



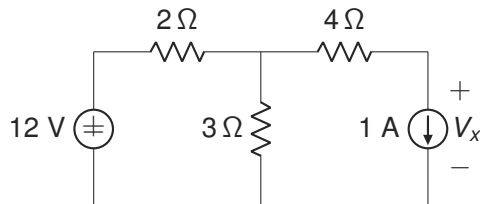
If a current source sits in only **one** mesh (an outside edge), that mesh current is known: $I_2 = \pm I_s$ (plus if it points the same way as I_2).

Don't write KVL for that mesh: the source's voltage is unknown.

Trap

The voltage across a current source is **not** zero. If asked for it, use KVL in its mesh **after** solving.

Try it: Example 6, current source on the edge



Find I_1 (left mesh, clockwise)
and the voltage V_x across the
current source.

your work here ↓

Example 6: solution

Known mesh the 1 A source is on mesh 2's outside edge, pointing down = clockwise for mesh 2: $I_2 = 1 \text{ A}$

Mesh 1 up the 12 V (+12), across the 2Ω ($-2I_1$), down the shared 3Ω ($-3(I_1 - 1)$): $12 - 2I_1 - 3I_1 + 3 = 0 \Rightarrow$

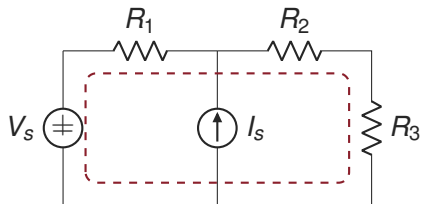
$$I_1 = 3 \text{ A}$$

V_x now walk mesh 2 clockwise, *including* the source's unknown voltage: across the 4Ω ($-4(1)$), down through the source entering its + ($-V_x$), up the 3Ω ($-3(1 - 3)$)

Solve $-4 - V_x + 6 = 0 \Rightarrow V_x = 2 \text{ V}$ (top +)

Shared current source: the supermesh

[NOT on formula sheet: memorize]



supermesh = outer loop

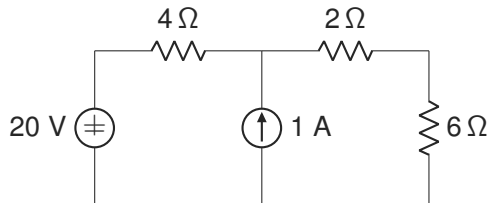
A current source in a **shared** branch:

1. Source equation: $i_2 - i_1 = I_s$ (the source's current is the difference).
2. One KVL around the **outside** of both meshes, skipping the source.

Trap

Getting the sign of $i_2 - i_1$ vs $i_1 - i_2$ wrong is the classic mistake. Ask: which mesh current flows the **same way** as the source arrow? That one is +.

Try it: Example 7, supermesh



The 1 A source points **up**.
Find both clockwise mesh currents.

your work here ↓

Example 7: solution

Source eq. up the middle = I_2 's direction (cw right mesh goes up its left side): $I_2 - I_1 = 1$

Outer KVL $20 - 4I_1 - 2I_2 - 6I_2 = 0$

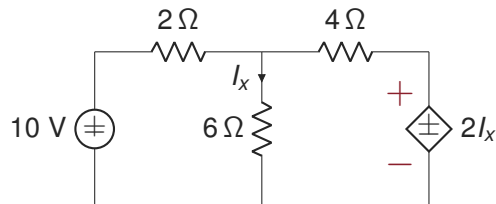
Substitute $20 - 4I_1 - 8(I_1 + 1) = 0 \Rightarrow 12I_1 = 12$

Result

$$I_1 = 1 \text{ A}, \quad I_2 = 2 \text{ A}$$

Check $20 - 4 - 4 - 12 = 0 \checkmark$

Try it: Example 8, mesh with a CCVS



I_x flows down through the $6\ \Omega$.

Find both clockwise mesh currents and I_x .

your work here ↓

Example 8: solution (1/2)

Plan KVL in both meshes, treating $2I_x$ like a known voltage source, then write I_x in mesh currents

Control I_x flows down the shared $6\ \Omega$, where the net current is $I_1 - I_2$:
 $I_x = I_1 - I_2$

Mesh 1 $10 - 2I_1 - 6(I_1 - I_2) = 0 \Rightarrow 8I_1 - 6I_2 = 10$

Example 8: solution (2/2)

Mesh 2 across the $4\ \Omega$ ($-4I_2$), down the $2I_x$ source entering its $+$ ($-2I_x$), up the $6\ \Omega$ ($-6(I_2 - I_1)$)

Substitute $-4I_2 - 2(I_1 - I_2) - 6I_2 + 6I_1 = 0 \Rightarrow 4I_1 - 8I_2 = 0$, so
 $I_1 = 2I_2$

Solve $16I_2 - 6I_2 = 10$:

$$I_2 = 1\text{ A}$$

$$I_1 = 2\text{ A}$$

$$I_x = 1\text{ A}$$

Nodal or mesh? Count the equations

Engineering shortcut: pick the method with fewer equations

Nodal: $(n - 1) - (\text{number of voltage sources})$

Mesh: $(\text{number of meshes}) - (\text{number of current sources})$

Many current sources: nodal is usually easier. Many voltage sources: mesh. Asked for currents: mesh gives them directly. Asked for voltages: nodal does.

Trap

If the exam says “use nodal” or “use mesh”, you **must** use that method, or risk 0 marks.

Why the counts work

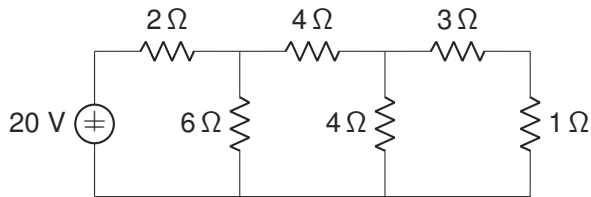
Nodal: start with $n - 1$ unknown node voltages. Each voltage source either fixes a node outright (to ground) or ties two nodes together (supernode). Either way it removes **one** unknown, so you need $(n - 1) - (\#V \text{ sources})$ equations.

Mesh: start with one unknown current per mesh. Each current source either fixes a mesh current (outside edge) or ties two mesh currents together (supermesh). Either way it removes **one** unknown: meshes $- \#I$ sources.

Example 9's circuit

5 nodes, 1 voltage source: nodal needs $4 - 1 = 3$. 3 meshes, 0 current sources: mesh needs 3. A tie, so pick whichever gives what's asked for (currents: mesh).

Try it: Example 9, three meshes (exam style)



Use mesh analysis.

(a) Write the 3 mesh equations.

(b) Find all mesh currents.

(c) Current in the $6\ \Omega$ and $4\ \Omega$ (vertical).

(d) Power from the source.

your work here ↓

Example 9: solution (1/2)

Matrix by inspection (all clockwise):

$$\begin{bmatrix} 8 & -6 & 0 \\ -6 & 14 & -4 \\ 0 & -4 & 8 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 0 \\ 0 \end{bmatrix}$$

Mesh 3 $-4I_2 + 8I_3 = 0 \Rightarrow I_2 = 2I_3$

Mesh 2 $-6I_1 + 28I_3 - 4I_3 = 0 \Rightarrow I_1 = 4I_3$

Mesh 1 $32I_3 - 12I_3 = 20 \Rightarrow I_3 = 1 \text{ A}, I_2 = 2 \text{ A}, I_1 = 4 \text{ A}$

Example 9: solution (2/2)

$$6\ \Omega \quad I_1 - I_2 = 2\text{ A down}$$

$$4\ \Omega \text{ (vertical)} \quad I_2 - I_3 = 1\text{ A down}$$

$$\text{Source} \quad P = 20 \times 4 = 80\text{ W supplied}$$

$$\text{Check} \quad 2(16) + 6(4) + 4(4) + 4(1) + 3(1) + 1(1) = 80\text{ W } \checkmark$$

Engineering shortcut: back-substitute from the far end

In a ladder, the last mesh equation only has two unknowns. Express everything in terms of the last mesh current, and the system collapses without a calculator.

Nodal traps

Check these before you box an answer

- Never write KCL at the reference node. It's redundant.
- Current source **into** the node goes on the right side as $+$. Out of the node is $-$.
- Node with a voltage source to ground: known voltage, no KCL there.
- Supernode: one KCL for the bubble, plus the source equation. Don't forget either one.
- The current through a voltage source is only found **after** solving.

Mesh traps

Check these before you box an answer

- Shared resistor: **difference** of mesh currents, own minus neighbour.
- Never walk KVL **through** a current source: its voltage is unknown. Use the source equation or a supermesh.
- Branch current $\neq$ mesh current on shared branches. Say which way it flows.
- Mesh analysis only works on flat (planar) circuits. If wires must cross, use nodal.
- Negative mesh current is fine: it just flows counterclockwise.

Exam-format traps

Use the method requested

“Use nodal” means nodal. The other method can get 0 marks.

Equations with variables first

Write the equations with R_1 , V_s , ... first, then plug in numbers. It's worth marks by itself.

Show the matrix

Writing the system in matrix form (even if you solve by substitution) shows your setup clearly and earns method marks.

Summary (1/2): nodal

Unknowns	node voltages, measured from the reference
Equations	KCL at each non-reference node: $\sum \frac{V_{\text{this}} - V_{\text{other}}}{R} = \text{source current entering}$
By inspection	diagonal = sum of G at the node, off-diagonal = $-G$ between nodes, RHS = current in
V source to ground	that node is known
Floating V source	supernode: one KCL + source equation
Dependent source	write the control in node voltages; matrix not symmetric

Summary (2/2): mesh and choosing

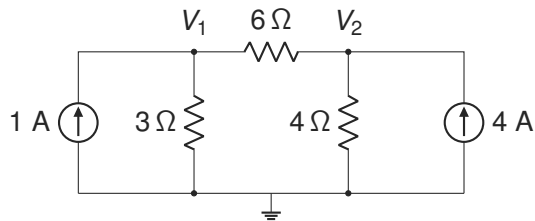
Unknowns	clockwise mesh currents
Equations	KVL around each mesh: shared R uses (own – neighbour)
By inspection	diagonal = sum of R in the mesh, off-diagonal = $-R$ shared, RHS = source rises
I source on edge	that mesh current is known
Shared I source	supermesh: source equation + outer KVL
Which method	nodal: $(n - 1) - \#V$ sources; mesh: meshes $-\#I$ sources. Fewer wins
Solving	clear fractions, Cramer for 2×2 , calculator for 3×3 , back-substitute ladders

Final check

8 problems. One per slide, with space. No looking back.

If one takes more than 10 min, go back to that section. The answer sheet comes after the last problem.

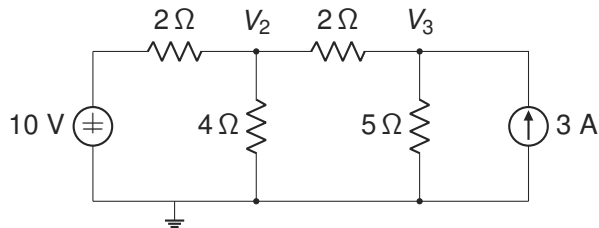
P1: nodal, two nodes



Use nodal analysis to find V_1 and V_2 .

your work here ↓

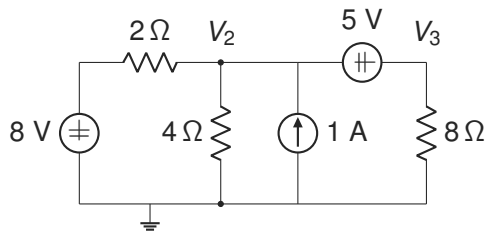
P2: nodal with a voltage source



Find V_2 , V_3 , and the current delivered by the 10 V source.

your work here ↓

P3: supernode

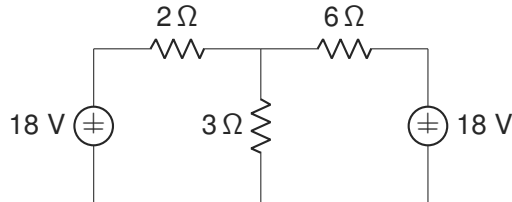


The 5 V source has + on the right.

Find V_2 and V_3 .

your work here ↓

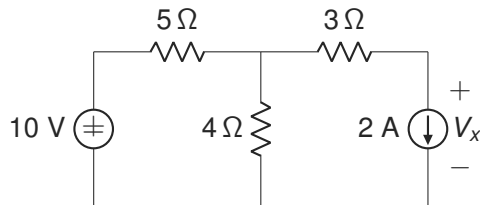
P4: two meshes



Both sources have + on top.
Find both clockwise mesh
currents and the current in the
 $3\ \Omega$ (with direction).

your work here ↓

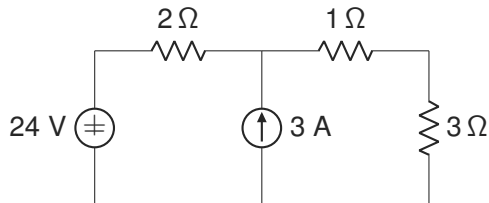
P5: current source on the edge



Find I_1 (left mesh), the current in the $4\ \Omega$, and V_x .

your work here ↓

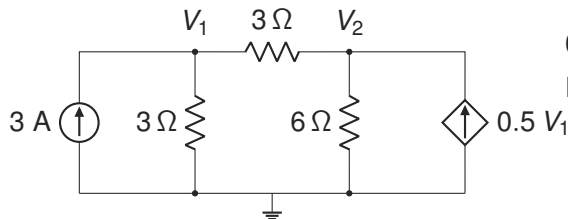
P6: supermesh



The 3 A source points **up**.
Find both clockwise mesh currents.

your work here ↓

P7: nodal with a VCCS

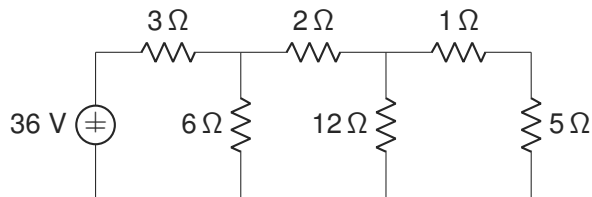


The dependent source pushes $0.5 V_1$ (A) **up** into V_2 .

Find V_1 and V_2 .

your work here ↓

P8: three meshes, total power



Use mesh analysis to find the total power dissipated in the circuit.

your work here ↓

Answer sheet

Full worked solutions to P1–P8.

P1: solution (1/2)

Plan two unknown node voltages and only current sources, so write KCL at V_1 and V_2 with every resistor current written as *leaving*

KCL V_1 leaving through $3\ \Omega$ and $6\ \Omega = 1\ \text{A}$ entering:

$$\frac{V_1}{3} + \frac{V_1 - V_2}{6} = 1$$

Clear $\times 6$: $2V_1 + V_1 - V_2 = 6 \Rightarrow 3V_1 - V_2 = 6$

KCL V_2 $\frac{V_2 - V_1}{6} + \frac{V_2}{4} = 4 \xrightarrow{\times 12} 2V_2 - 2V_1 + 3V_2 = 48 \Rightarrow$
 $-2V_1 + 5V_2 = 48$

P1: solution (2/2)

Substitute from the first: $V_2 = 3V_1 - 6$, so $-2V_1 + 15V_1 - 30 = 48$
 $\Rightarrow 13V_1 = 78$

Result

$$V_1 = 6 \text{ V}$$

$$V_2 = 3(6) - 6 = 12 \text{ V}$$

Check

at V_1 : $2 + (-1) = 1 \text{ A}$ ✓ at V_2 : $1 + 3 = 4 \text{ A}$ ✓

P2: solution

Plan V_1 is fixed by the source (10 V to ground), so only V_2 and V_3 are unknown: two KCL equations

$$\text{KCL } V_2 \quad \frac{V_2 - 10}{2} + \frac{V_2}{4} + \frac{V_2 - V_3}{2} = 0 \xrightarrow{\times 4} 5V_2 - 2V_3 = 20$$

$$\text{KCL } V_3 \quad \frac{V_3 - V_2}{2} + \frac{V_3}{5} = 3 \xrightarrow{\times 10} -5V_2 + 7V_3 = 30$$

Eliminate add the two equations: $5V_3 = 50 \Rightarrow V_3 = 10 \text{ V}$

Back $5V_2 = 20 + 2(10) = 40 \Rightarrow V_2 = 8 \text{ V}$

Source / the source's current is the 2Ω current: $(10 - 8)/2 =$
 $1 \text{ A out of the } +$

P3: solution (1/2)

Plan $V_1 = 8\text{ V}$ is known. The 5 V source floats between V_2 and V_3 : supernode (one KCL for both, plus the source equation)

Source eq. + on the V_3 side: $V_3 = V_2 + 5$

Supernode all currents leaving the bubble:
$$\frac{V_2 - 8}{2} + \frac{V_2}{4} - 1 + \frac{V_3}{8} = 0 \quad (\text{the } 1\text{ A enters, so } -1)$$

P3: solution (2/2)

Substitute $\times 8$ and $V_3 = V_2 + 5$: $4V_2 - 32 + 2V_2 - 8 + V_2 + 5 = 0$
 $\Rightarrow 7V_2 = 35$

Result

$$V_2 = 5 \text{ V}$$

$$V_3 = 10 \text{ V}$$

Check $-1.5 + 1.25 - 1 + 1.25 = 0 \checkmark$

P4: solution (1/2)

Plan two voltage sources, no current sources: plain mesh analysis, written with the by-inspection matrix (diagonal = resistance around each mesh, off-diagonal = $-$ shared resistance)

Matrix
$$\begin{bmatrix} 5 & -3 \\ -3 & 9 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 18 \\ -18 \end{bmatrix}$$

Cramer $\det = 45 - 9 = 36, \quad I_1 = \frac{162 - 54}{36}, \quad I_2 = \frac{-90 + 54}{36}$

P4: solution (2/2)

Result $I_1 = 3 \text{ A}$, $I_2 = -1 \text{ A}$ (1 A counterclockwise)

$$3 \Omega \quad I_1 - I_2 = 3 - (-1) = 4 \text{ A downward}$$

Trap

$I_1 - I_2$ with a negative I_2 : the minus signs add. $3 - (-1) = 4$, not 2.

P5: solution (1/2)

Plan the 2 A source is on the outside edge of mesh 2, so I_2 is known. Only mesh 1 needs KVL. Then use mesh 2's KVL to get V_x

Known the arrow points down the right edge, which is clockwise for mesh 2: $I_2 = 2 \text{ A}$

Mesh 1 $10 - 5I_1 - 4(I_1 - I_2) = 0 \Rightarrow 10 - 5I_1 - 4I_1 + 8 = 0 \Rightarrow$
 $I_1 = 2 \text{ A}$

P5: solution (2/2)

4Ω shared branch: $I_1 - I_2 = 2 - 2 = 0\text{ A}$

Mesh 2 walk clockwise: $-3I_2 - V_x - 4(I_2 - I_1) = 0 \Rightarrow$
 $-6 - V_x - 0 = 0 \Rightarrow V_x = -6\text{ V}$

Meaning negative: the real + of the current source is at the **bottom**,
6 V

P6: solution (1/2)

Plan the 3 A source sits in the **shared** branch: supermesh
(source equation + one KVL around the outside)

Source eq. the arrow points up the middle. Clockwise I_2 goes up its left side, clockwise I_1 goes down its right side, so
 $I_2 - I_1 = 3$

Outer KVL skip the source branch and walk the outside:
 $24 - 2I_1 - 1 I_2 - 3I_2 = 0$

P6: solution (2/2)

Substitute $I_2 = I_1 + 3$: $24 - 2I_1 - 4I_1 - 12 = 0 \Rightarrow 6I_1 = 12$

Result $I_1 = 2 \text{ A}$, $I_2 = 5 \text{ A}$

Check $24 - 4 - 5 - 15 = 0 \checkmark$

P7: solution

Plan KCL at both nodes, then replace the control variable V_1 (it's already an unknown, so no extra equation is needed)

$$\text{KCL } V_1 \quad \frac{V_1}{3} + \frac{V_1 - V_2}{3} = 3 \Rightarrow 2V_1 - V_2 = 9$$

$$\text{KCL } V_2 \quad \frac{V_2 - V_1}{3} + \frac{V_2}{6} = 0.5 V_1 \xrightarrow{\times 6} 3V_2 = 5V_1$$

$$\text{Substitute} \quad 2V_1 - \frac{5}{3}V_1 = 9 \Rightarrow \frac{1}{3}V_1 = 9$$

Result

$$V_1 = 27 \text{ V}, \quad V_2 = 45 \text{ V}$$

$$\text{Check } V_2 \quad \frac{45-27}{3} + \frac{45}{6} = 6 + 7.5 = 13.5 = 0.5(27) \checkmark$$

P8: solution (1/2)

Plan total power dissipated = total power supplied by the one source, so we only need I_1 . Solve the ladder by back-substituting from the last mesh

Matrix
$$\begin{bmatrix} 9 & -6 & 0 \\ -6 & 20 & -12 \\ 0 & -12 & 18 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 36 \\ 0 \\ 0 \end{bmatrix}$$

Back-sub mesh 3: $I_2 = 1.5 I_3$; mesh 2: $I_1 = 3 I_3$; mesh 1: $18 I_3 = 36$

P8: solution (2/2)

Currents $I_3 = 2 \text{ A}$, $I_2 = 3 \text{ A}$, $I_1 = 6 \text{ A}$

Total power = power supplied = $36 \times 6 = 216 \text{ W}$

Check $108 + 54 + 18 + 12 + 4 + 20 = 216 \text{ W} \checkmark$